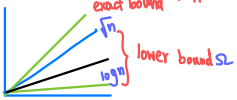
If f(n) = log2(n) and g(n) = sqrt(n), which one below is true? * 1 point

- f(n) is Omega(g(n))
- f(n) is O(g(n))
- f(n) is Theta(g(n))
- g(n) is O(f(n))
- g(n) is Theta(f(n))

f(n) = log2(n) < g(n) = $\sqrt{n}$

f(n) = O(g(n)) log2 = ↑ upper bound O




3. What is the worst-case efficiency of root key deletion from a heap? * 1 point

worst case : O(n log n)

- Theta(n)
- Theta(log2 n)
- Theta(n log2 n)
- Theta(n^2)
- Theta(n + log2 n)

Selection sort

Radix sort

4. Suppose we want to sort a list of students by their exam scores, where the input list is arranged by student IDs, and the sorted output preserves the order of student IDs in the input list for duplicate scores, which algorithm below is not suitable for the task? * 1 point

| NO. | Score | | |
|---|---|---|---|
| 1 | 6 | 2 | 7 |
| 2 | 7 | 1 | 6 |
| 3 | 6 | 3 | 6 |
| 4 | 3 | 4 | 3 |

same เรียงจาก ID

Heap sort

Insertion sort — high efficiency คุณ.เขตละช้า ถ้าข้อมูลที่ overload

Bubble sort

Selection sort

Merge sort — divide - and - conquer

5. For a list of fixed-length keys such as XI7FS6, PL4ZQ2, JI8FR9, XL8FQ6, PY2ZR5, KV7WS9, JL2ZV3, KI4WR2, which sorting algorithm below is * 1 point

- Bubble sort
- Selection sort
- Merge sort

5. For a list of fixed-length keys such as XI7FS6, PL4ZQ2, JI8FR9, XL8FQ6, PY2ZR5, KV7WS9, JL2ZV3, KI4WR2, which sorting algorithm below is appropriate to sort such input in terms of time efficiency? * 1 point

- (✓) Radix sort
- Quick sort
- Merge sort
- Insertion sort
- Heap sort

6. Which one below is correct about expressing an algorithm in programming languages ? * 1 point

## Part 2 (6 questions, 14 points):

1. (3 points) Given a list {9, 12, 5, 30, 17, 20, 8, 4}, what is the result of Hoare partition? * 3 points

- {8,4,5}, 9, {20,17,30,12}
- {4,8,5}, 9, {17,12,30,20}
- {8,4,5}, 9, {17,20,30,12}
- {4,5,8}, 9, {17,20,12,30}
- {8,4,5}, 9, {30,20,17,12}
- None is correct

2. A sequence {9,6,8,2,5,7} is the array representation of the heap. * 1 point

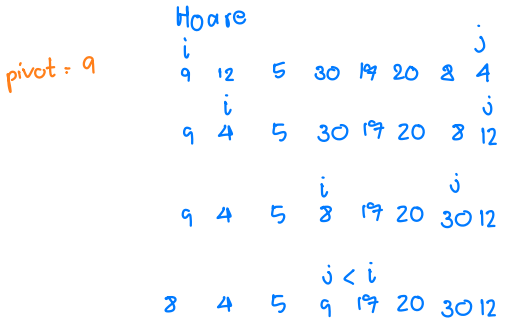
- True
- False

3. (2 points) How many key comparisons to sort the sequence {'A', 'L', 'G', 'O', 'R', 'I', 'T', 'H', 'M'} alphabetically by using Insertion sort? * 2 points

- 9
- 15
- 19
- 21
- 25
- None is correct.

(1)



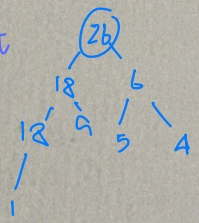

**Algorithm** HoarePartition

1: Input: Array $A[l \dots r]$ of orderable elements
2: Output: A partition of $A[l \dots r]$ with the split position as return value
3:
4: $p \leftarrow A[l]$ ← Take the pivot element
5: $i \leftarrow l$
6: $j \leftarrow r + 1$
7: **repeat**
8: **repeat** $i \leftarrow i + 1$ **until** $A[i] \geq p$
9: **repeat** $j \leftarrow j - 1$ **until** $A[j] \leq p$
10: swap$(A[i], A[j])$
11: **until** $i \geq j$ ↳ Terminate if $i \geq j$
12: swap$(A[i], A[j])$ ▷ Undo last swap when $i \geq j$
13: swap$(A[l], A[j])$
14: **return** $j$



4. (2 points) Suppose we were to construct a heap from the input sequence {1,6,26,9,18,5,4,18} by using the top-down heap construction, what is the key in the last leaf node in the heap? * 2 points

Max node → root




- 6
- 9
- 5
- 4
- 1

5. (3 points) Suppose a heap sort is applied to sort the input sequence {1,6,26,9,18,5,4,18}. The sorted output is stable. * 3 points

Not stable  Heap Select Quick

- True
- False

Insertion sort

Heap sort

6. Which one below is correct about expressing an algorithm in programming languages ? * 1 point

- Ideas are conveyed more clearly than natural languages or pseudo codes. ✓
- It is easiest to understand. ✓
- Instructions are unambiguous. ✓
- Meanings of syntax are known to most people. ✓
- All of them is correct.

7. Which one is incorrect about measuring algorithm efficiency by using counts of basic operations ? * 1 point

- It is independent of programming languages implemented.

8. The growth order of algorithm running time tells the amount of time taken by an algorithm to solve the problem of a given input size n. * 1 point

- [ ] True
- [x] False

9. If you want to express a lower bound of algorithm efficiency, which notation you have to use ? * 1 point

- [ ] Big Oh
- [ ] Big Theta
- [x] Big Omega
- [ ] Big Alpha
- [ ] Big Gamma

10. An algorithm with growth order log2(n) has higher running time than the one with growth order sqrt(n) when the input size n grows extremely large. * 1 point

- [ ] True
- [x] False

- True
- False

9. If you want to express a lower bound of algorithm efficiency, which notation you have to use ? * 1 point

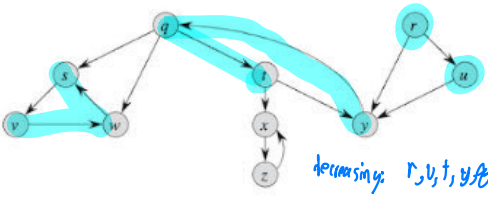
- Big Oh
- Big Theta
- Big Omega
- Big Alpha
- Big Gamma

10. An algorithm with growth order log2(n) has higher running time than the one with growth order sqrt(n) when the input size n grows extremely large. * 1 point

- True

node t (break ties alphabetically), node w will be in which SCC based on the order of SCCs produced by the algorithm?

Kosaraju's

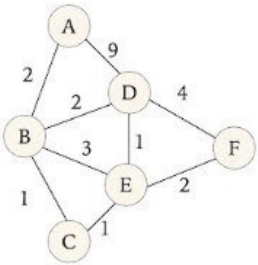



decreasing: r, u, t, y, q, s, v, w, x, z

- First SSC
- Second SSC
- Third SSC
- Fourth SSC
- Fifth SSC

w 8, 3
v 7, 4
s 6, 5
q 5, 6
y 4, 7
z 3, 1
x 2, 2
t 1, 8
u 10, 9
r 9, 10

From the graph below, the minimum spanning tree is the same as the shortest path tree rooted at node A. * 2 points




- True
- False

Given a list {9, 12, 5, 30, 17, 20, 8, 4}, what is the result of Lumoto partition? * 3 points

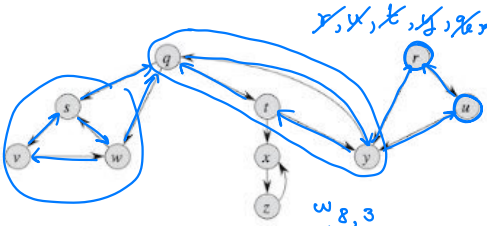
- {8,4,5}, 9, {20,17,30,12}
- {4,8,5}, 9, {17,12,30,20}
- {8,4,5}, 9, {17,20,30,12}
- {4,5,8}, 9, {17,20,12,30}
- {8,4,5}, 9, {30,20,17,12}

(1)

Lomuto

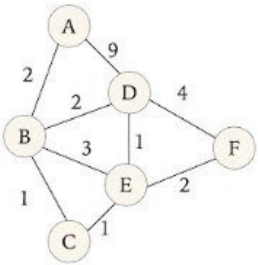
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| | l | | | | | | | r |
| pivot=9 | s 9 | 12 | i 5 | 30 | 17 | 20 | 8 | 4 |
| | 9 | 5 | 8 | S 4 | 17 | 20 | 12 | t 30 |
| swap(1,5) | 4 | 5 | 8 | 9 | 17 | 20 | 12 | 30 |

node t (break ties alphabetically), node w will be in which SCC based on the order of SCCs produced by the algorithm?




- [ ] First SSC
- [x] Second SSC
- [ ] Third SSC
- [ ] Fourth SSC
- [ ] Fifth SSC

From the graph below, the minimum spanning tree is the same as the shortest path tree rooted at node A. * 2 points




- [ ] True
- [x] False

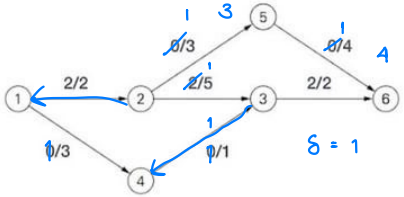
Given a list {9, 12, 5, 30, 17, 20, 8, 4}, what is the result of Lumoto partition? * 3 points

- [ ] {8,4,5}, 9, {20,17,30,12}
- [ ] {4,8,5}, 9, {17,12,30,20}
- [ ] {8,4,5}, 9, {17,20,30,12}
- [x] {4,5,8}, 9, {17,20,12,30}
- [ ] {8,4,5}, 9, {30,20,17,12}

09:49 Fri 28 Oct
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- [x] Designing a transmission network with smallest total link cost.
- [ ] Finding a largest cost path between any two nodes in a graph.

For the following flow network with source node 1 and destination node 6, which one is a valid augmenting path in the residual graph ? * 1 คะแนน

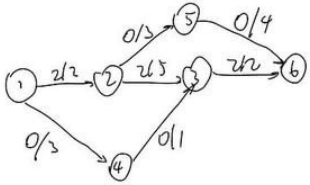


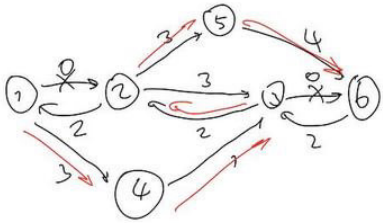

- 1 -> 2 -> 3 -> 4
- 1 -> 4 -> 3 -> 2 -> 5 -> 6
- 1-> 2 -> 5 -> 6
- 1 -> 4 -> 3 -> 6

หน้า 2 จาก 3

กลับ ถัดไป ล้างแบบฟอร์ม

ห้ามส่งรหัสผ่านใน Google ฟอร์ม
แบบฟอร์มนี้ถูกสร้างขึ้นภายใน King Mongkut's University of Technology Thonburi รายงานการ



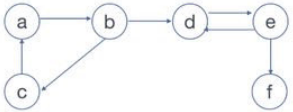



- [ ] C
- [ ] D
- [ ] E
- [ ] F

An edge in a flow network has the edge capacity of 16 units and a flow of 5 units. What are the capacities of the forward edge and the reverse edge in the residual graph ? * 2 points

- ( ) Forward edge capacity of 5 units and Reverse edge capacity of 5 units
- ( ) Forward edge capacity of 16 units and Reverse edge capacity of 11 units
- (•) Forward edge capacity of 11 units and Reverse edge capacity of 5 units
- ( ) Forward edge capacity of 11 units and Reverse edge capacity of 16 units
- ( ) Forward edge capacity of 5 units and Reverse edge capacity of 11 units

How many strongly connected components are there in the graph below? * 1 point



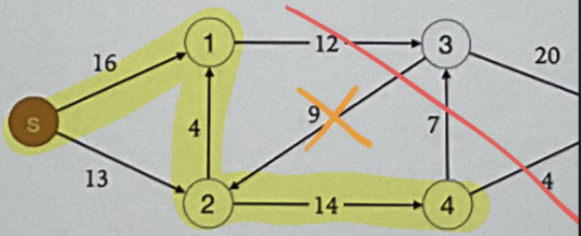

- ( ) 2
- (•) 3
- ( ) 4

Voice Connected: business meeting / Tabbytronic Supercomp...

cheggy

PartC

For the following s-t flow network, what is the minimum cu

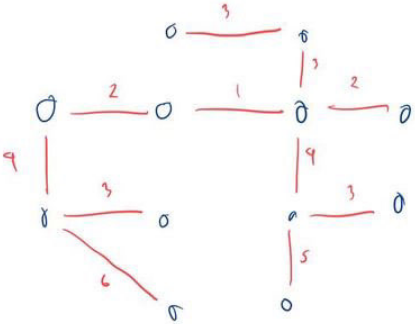



- {s, 1}
- {s, 1, 2}
- {s, 1, 3}
- {s, 1, 2, 4}
- {s, 1, 3, 4}

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PartC

t is the cost of minimum spanning tree of the g



3+3+2+1+2+4+4+7+8+6+5

6 + 5 + 8 + 6 + 11

11 + 14 + 11 = 22+14 = 36

5 of 5

Drywall

Buy Appliances

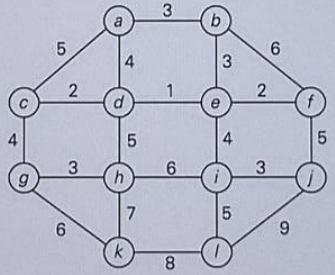
Obtain Permit

Wiring

Initial inspection

What is the cost of minimum spanning tree of the graph below ? *

2 points




28

31

33

36

38

41

If we run Depth first search by starting at node A, and then the subsequent

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Quiz 2: PHY1...
Self Test 2-1...

1. MaxElement() is an O(n) algorithm. (1 Point) *

- [x] True
- [ ] False

2. MaxElement() is Theta(1) in best case. (1 Point) *

- [x] True
- [ ] False

3. MaxElement() is Theta(n) in worst case. (1 Point) *

- [x] True
- [ ] False

4. MaxElement() is O(n) in all cases. (1 Point) *

- [ ] True
- [x] False

5. An Theta(n) algorithm runs faster than Theta(n^2) algorithm. (1 Point) *

- [x] True
- [ ] False

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เอาเพลงไรดีๆ#ตามจังหวะ#2รูป#ทรงเชง
now
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Travel Town...
Quiz 2: PHY1...
PORNCHAN...
Can't Open...
Self Test 2-2...

1. What is the value M(n) of the recursive Tower of Hanoi algorithm ? (1 Point)

- n-1
- n^2
- 2^n
- 2^{n+1}
- 2^n - 1 (selected)

2. What is the efficiency of the recursive Tower of Hanoi algorithm? (1 Point)

- Theta(n)
- Theta(n log2 n)
- Theta(n^2) (selected)
- Theta(2^n)
- No idea

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